

SAMRUDHI BADAKH

Computer Science Undergraduate | COEP Technological University

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PROFESSIONAL SUMMARY

Computer Science undergraduate at COEP Technological University with hands-on experience across embedded systems, computer vision, and full-stack web development. Contributes to a live satellite project (Communication Software team) implementing low-level hardware drivers in C. Holds the Google AI Essentials Specialization and Google Cloud badges in Generative AI and Prompt Engineering. Actively building AI/ML skills with 150+ LeetCode problems solved. Seeking software or embedded development internships to grow in industry.

EDUCATION

COEP Technological University – Pune, Maharashtra

Bachelor of Technology – Computer Science & Engineering

2024 – Present

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, SQL

Embedded & Hardware: HSMCI & SPI Protocols, ARM Cortex-M4, SAM4S Microcontroller, Register-level C

AI / ML, CV & GenAI: OpenCV, face_recognition (dlib), NumPy, Prompt Engineering (Google certified); exploring CNNs and deep learning

Web Technologies: HTML, CSS, JavaScript, React, Three.js, Node.js, Express.js, FastAPI, PostgreSQL

Developer Tools: Git, GitHub, GitLab, VS Code, Linux

Core Competencies: Data Structures & Algorithms (150+ LeetCode), OOP, REST API Design, Debugging

PROJECTS

SD Card Interface via HSMCI & SPI – SAM4S Microcontroller | *University Satellite Club*

- Implemented SD card communication on the SAM4S ARM Cortex-M4 using both HSMCI and SPI peripherals in Embedded C
- Handled low-level hardware interfacing including register configuration, command/response protocol, and embedded debugging
- Part of the Communication Software sub-team; contributes to onboard data storage for the university's active satellite project

Face Detection & Recognition System

- Built a real-time face detection and recognition pipeline in Python using OpenCV and the face_recognition library (dlib-based deep metric learning)
- Implemented face encoding, multi-face tracking across video frames, and identity matching against a known-faces database
- Exploring CNN-based approaches to improve recognition robustness under varying lighting, angles, and occlusion

Portfolio Website | *samrudhibadakh.netlify.app*

- Designed and built a full-stack personal portfolio using Node.js, Express.js, and PostgreSQL, deployed on Netlify
- Showcases projects, skills, and resume with a responsive UI built in HTML, CSS, and JavaScript

Three.js CSAT Website | csat-web.web.app

- Built an interactive 3D website for the university satellite club using Three.js and vanilla JavaScript
- Implemented custom 3D scene management, camera controls, and animations for an immersive web experience

Todo REST API – FastAPI

- Built a RESTful task management API using Python and FastAPI with full CRUD operations and Swagger/OpenAPI documentation
- Implemented request validation with Pydantic models; hands-on experience with REST design and Python async patterns

CERTIFICATIONS

Google AI Essentials Specialization (5 courses) – Google / Coursera | May 2026

- Introduction to AI [Verify](#)
- Maximize Productivity With AI Tools [Verify](#)
- Discover the Art of Prompting [Verify](#)
- Use AI Responsibly [Verify](#)
- Stay Ahead of the AI Curve [Verify](#)
- Start Writing Prompts like a Pro [Verify](#)

Google Cloud Skill Badges

- Introduction to Generative AI [View Badge](#)
- Prompt Design in Agent Platform [View Badge](#)
- Responsible AI: Applying AI Principles with Google Cloud [View Badge](#)

ACHIEVEMENTS & ACTIVITIES

- Qualified JEE Advanced 2024
- 97.6 percentile in JEE Mains 2024 (top 2.4% nationwide out of ~1.2 million candidates)
- 97.83% in CBSE Class 10th | 84.33% in HSC Class 12th
- 150+ problems solved on LeetCode (arrays, strings, DP, graphs, trees)
- Active member: University Satellite Club (Communication Software), Table Tennis Team, Basketball Team